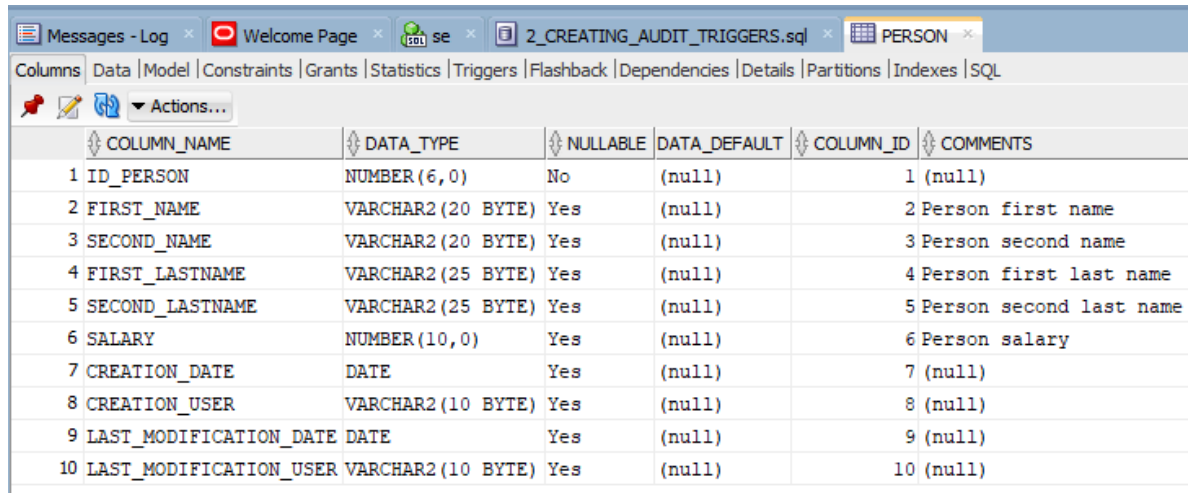


Evidences

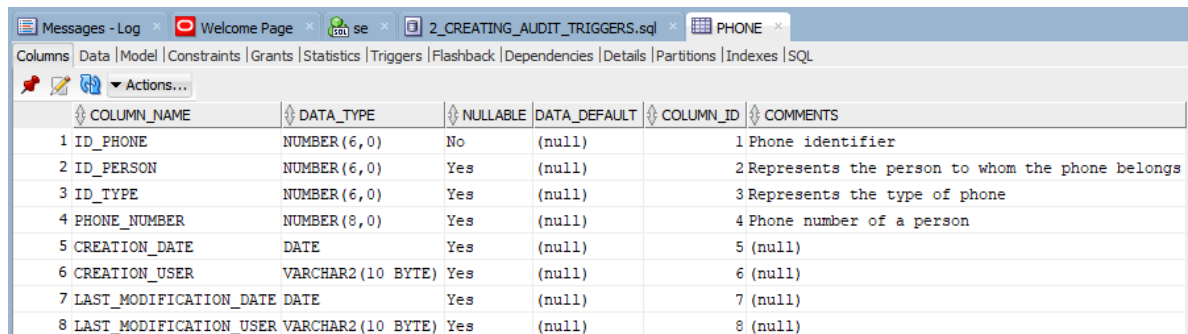
*The logical and conceptual model of the previous lab was used, also they were not attached because the teacher indicated that it was not necessary.

1. Create the audit fields



The screenshot shows the 'PERSON' table structure in SQL Developer. The table has 10 columns: ID_PERSON, FIRST_NAME, SECOND_NAME, FIRST_LASTNAME, SECOND_LASTNAME, SALARY, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. Each column's data type, nullability, default value, and comments are listed.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_PERSON	NUMBER(6,0)	No	(null)	1	(null)
2	FIRST_NAME	VARCHAR2(20 BYTE)	Yes	(null)	2	Person first name
3	SECOND_NAME	VARCHAR2(20 BYTE)	Yes	(null)	3	Person second name
4	FIRST_LASTNAME	VARCHAR2(25 BYTE)	Yes	(null)	4	Person first last name
5	SECOND_LASTNAME	VARCHAR2(25 BYTE)	Yes	(null)	5	Person second last name
6	SALARY	NUMBER(10,0)	Yes	(null)	6	Person salary
7	CREATION_DATE	DATE	Yes	(null)	7	(null)
8	CREATION_USER	VARCHAR2(10 BYTE)	Yes	(null)	8	(null)
9	LAST_MODIFICATION_DATE	DATE	Yes	(null)	9	(null)
10	LAST_MODIFICATION_USER	VARCHAR2(10 BYTE)	Yes	(null)	10	(null)

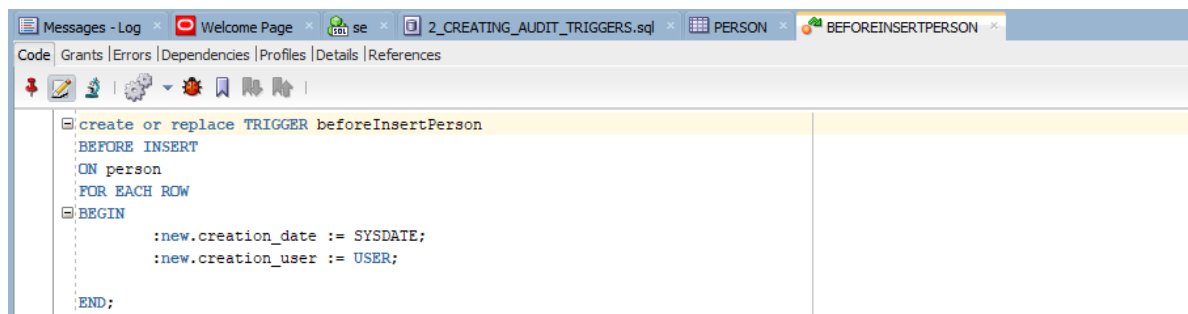


The screenshot shows the 'PHONE' table structure in SQL Developer. The table has 8 columns: ID_PHONE, ID_PERSON, ID_TYPE, PHONE_NUMBER, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. Each column's data type, nullability, default value, and comments are listed.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_PHONE	NUMBER(6,0)	No	(null)	1	Phone identifier
2	ID_PERSON	NUMBER(6,0)	Yes	(null)	2	Represents the person to whom the phone belongs
3	ID_TYPE	NUMBER(6,0)	Yes	(null)	3	Represents the type of phone
4	PHONE_NUMBER	NUMBER(8,0)	Yes	(null)	4	Phone number of a person
5	CREATION_DATE	DATE	Yes	(null)	5	(null)
6	CREATION_USER	VARCHAR2(10 BYTE)	Yes	(null)	6	(null)
7	LAST_MODIFICATION_DATE	DATE	Yes	(null)	7	(null)
8	LAST_MODIFICATION_USER	VARCHAR2(10 BYTE)	Yes	(null)	8	(null)

2.

Trigger:



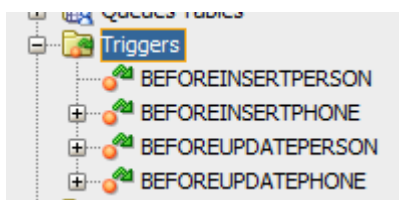
The screenshot shows the SQL code for a trigger named 'beforeInsertPerson' in SQL Developer. The trigger is created or replaced, is a BEFORE INSERT trigger on the 'person' table, and is triggered for each row. The trigger body sets the 'creation_date' to the current system date (SYSDATE) and the 'creation_user' to the current user (USER).

```
create or replace TRIGGER beforeInsertPerson
BEFORE INSERT
ON person
FOR EACH ROW
BEGIN
    :new.creation_date := SYSDATE;
    :new.creation_user := USER;
END;
```

```
create or replace TRIGGER beforeInsertPhone
BEFORE INSERT
ON phone
FOR EACH ROW
BEGIN
    :new.creation_date := SYSDATE;
    :new.creation_user := USER;
END;
```

```
create or replace TRIGGER beforeUpdatePerson
BEFORE UPDATE
ON person
FOR EACH ROW
BEGIN
    :new.last_modification_date := SYSDATE;
    :new.last_modification_user := USER;
END;
```

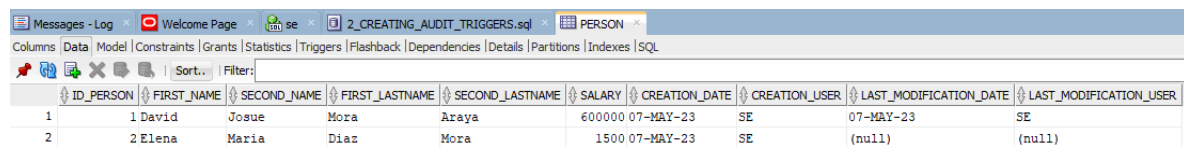
```
create or replace TRIGGER beforeUpdatePhone
BEFORE UPDATE
ON phone
FOR EACH ROW
BEGIN
    :new.last_modification_date := SYSDATE;
    :new.last_modification_user := USER;
END;
```



Inserts person:

ID_PERSON	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME	SALARY	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	Diego	Josue	Mora	Araya	600000	07-MAY-23	SE	(null)	(null)
2	Elena	Maria	Diaz	Mora	1500	07-MAY-23	SE	(null)	(null)

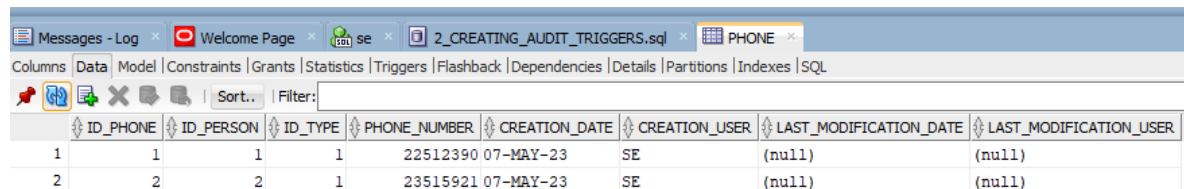
Update person 1:



The screenshot shows the SQL Developer interface with the PERSON table selected. The table has columns: ID_PERSON, FIRST_NAME, SECOND_NAME, FIRST_LASTNAME, SECOND_LASTNAME, SALARY, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. There are two rows of data.

ID_PERSON	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME	SALARY	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	David	Josue	Mora	Araya	600000	07-MAY-23	SE	07-MAY-23	SE
2	Elena	Maria	Diaz	Mora	1500	07-MAY-23	SE	(null)	(null)

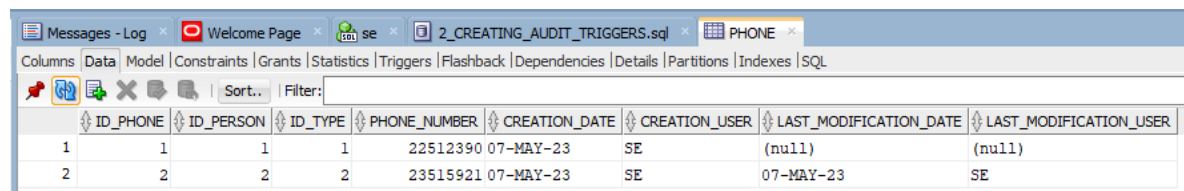
Insert phone:



The screenshot shows the SQL Developer interface with the PHONE table selected. The table has columns: ID_PHONE, ID_PERSON, ID_TYPE, PHONE_NUMBER, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. There are two rows of data.

ID_PHONE	ID_PERSON	ID_TYPE	PHONE_NUMBER	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	1	1	22512390	07-MAY-23	SE	(null)	(null)
2	2	2	23515921	07-MAY-23	SE	(null)	(null)

Update phone 2:

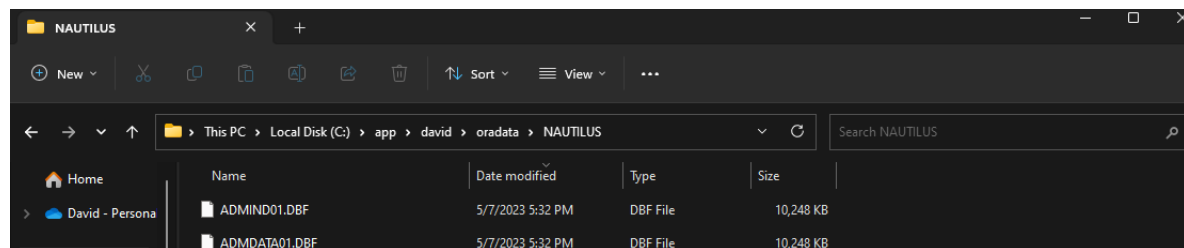


The screenshot shows the SQL Developer interface with the PHONE table selected. The table has columns: ID_PHONE, ID_PERSON, ID_TYPE, PHONE_NUMBER, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. There are two rows of data.

ID_PHONE	ID_PERSON	ID_TYPE	PHONE_NUMBER	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	1	1	22512390	07-MAY-23	SE	(null)	(null)
2	2	2	23515921	07-MAY-23	SE	07-MAY-23	SE

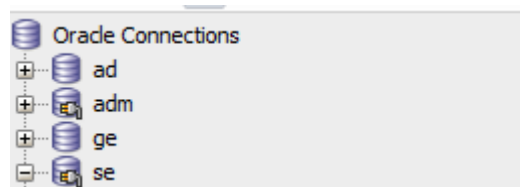
3.

ADM Tablespace



The screenshot shows a Windows File Explorer window titled 'NAUTILUS'. The address bar shows the path: This PC > Local Disk (C:) > app > david > oradata > NAUTILUS. The search bar contains 'Search NAUTILUS'. The file list shows two files: ADMIN01.DBF and ADMDATA01.DBF, both of type 'DBF File' and size '10,248 KB', with a date modified of '5/7/2023 5:32 PM'.

Name	Date modified	Type	Size
ADMIN01.DBF	5/7/2023 5:32 PM	DBF File	10,248 KB
ADMDATA01.DBF	5/7/2023 5:32 PM	DBF File	10,248 KB



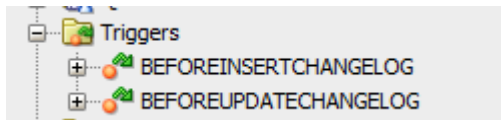
4.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	SCHEMA_NAME	VARCHAR2(20 BYTE)	Yes	(null)	1 (null)	
2	TABLE_NAME	VARCHAR2(20 BYTE)	Yes	(null)	2 (null)	
3	FIELD_NAME	VARCHAR2(20 BYTE)	Yes	(null)	3 (null)	
4	PREVIOUS_VALUE	NUMBER(10,0)	Yes	(null)	4 (null)	
5	CURRENT_VALUE	NUMBER(10,0)	Yes	(null)	5 (null)	

	CONSTRAINT_NAME	CONSTRAINT_TYPE	SEARCH_CONDITION
1	CHANGELOG_CURRENT_VALUE_NN	Check	current_value is NOT NULL
2	CHANGELOG_FIELD_NAME_NN	Check	field_name is NOT NULL
3	CHANGELOG_PREVIOUS_VALUE_NN	Check	previous_value is NOT NULL
4	CHANGELOG_SCHEMA_NAME_NN	Check	schema_name is NOT NULL
5	CHANGELOG_TABLE_NAME_NN	Check	table_name is NOT NULL

5.

Triggers ADM



Procedure beforeInsertUpdateSalary on sheme se

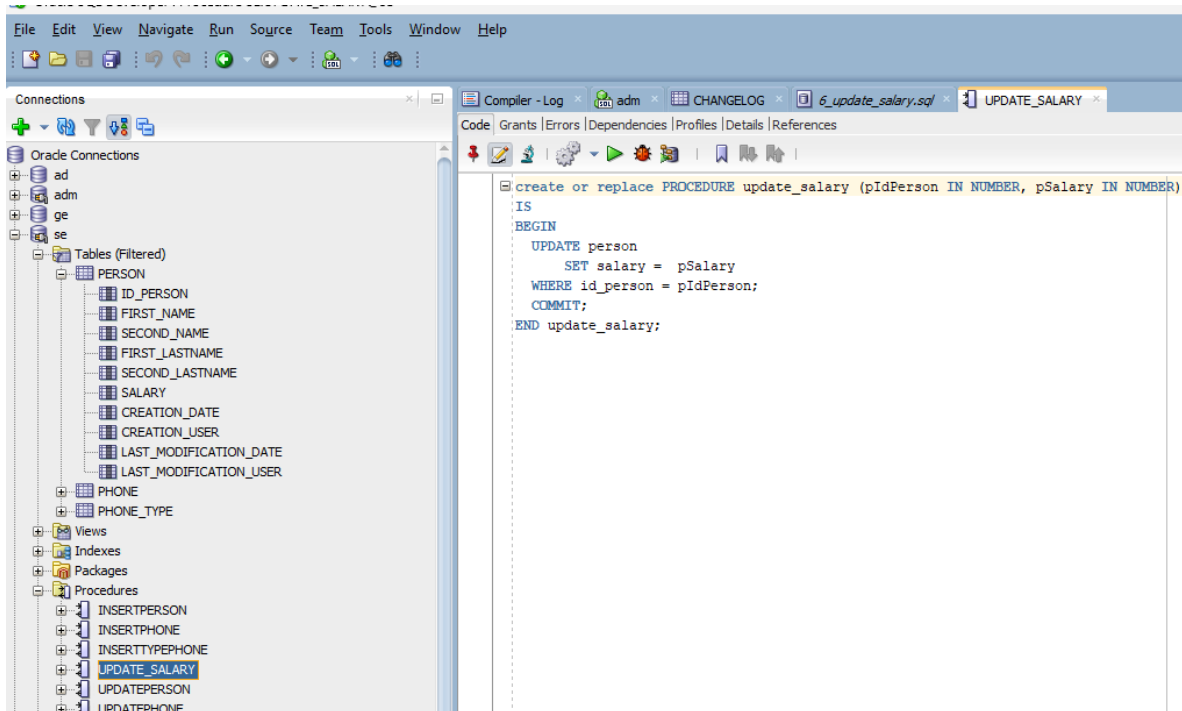
```

create or replace TRIGGER beforeInsertUpdateSalary
BEFORE INSERT OR UPDATE
ON salary
ON person
FOR EACH ROW
BEGIN
    IF UPDATING THEN
        adm.insertChangelog('SE','PERSON','SALARY',:OLD.last_modification_date, :OLD.salary, :NEW.salary, :NEW.creation_date,:NEW.last_modification_date);
    END IF;
END beforeInsertUpdateSalary;

```

6.

Procedure update_salary



Before update

The screenshot shows the 'PERSON' table data in Oracle SQL Developer. The table has 10 columns: ID_PERSON, FIRST_NAME, SECOND_NAME, FIRST_LASTNAME, SECOND_LASTNAME, SALARY, CREATION_DATE, CREATION_USER, LAST_MODIFICATION_DATE, and LAST_MODIFICATION_USER. The data is as follows:

ID_PERSON	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME	SALARY	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	David	Josue	Mora	Araya	600000	07-MAY-23	SE	07-MAY-23	SE
2	Elena	Maria	Diaz	Mora	1500	07-MAY-23	SE	(null)	(null)

After update

The screenshot shows the 'PERSON' table data in Oracle SQL Developer after the update. The data is as follows:

ID_PERSON	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME	SALARY	CREATION_DATE	CREATION_USER	LAST_MODIFICATION_DATE	LAST_MODIFICATION_USER
1	David	Josue	Mora	Araya	600000	07-MAY-23	SE	07-MAY-23	SE
2	Elena	Maria	Diaz	Mora	2000	07-MAY-23	SE	07-MAY-23	SE

After update on changelog

The screenshot shows the 'CHANGELOG' table data in Oracle SQL Developer. The table has 8 columns: SCHEMA_NAME, TABLE_NAME, FIELD_NAME, PREVIOUS_VALUE, CURRENT_VALUE, CREATION_DATE, UPDATE_DATE, and CREATION_USER. The data is as follows:

SCHEMA_NAME	TABLE_NAME	FIELD_NAME	PREVIOUS_VALUE	CURRENT_VALUE	CREATION_DATE	UPDATE_DATE	CREATION_USER
SE	PERSON	SALARY	1500	2000	07-MAY-23	07-MAY-23	SE